Agentic AI-Based Dynamic Tariff Optimization for EV Charging Networks

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This project presents an intelligent EV charging tariff optimization framework using Agentic AI techniques to dynamically regulate electricity pricing, reduce congestion, improve utilization efficiency, and increase revenue performance across EV charging networks. The system analyzes charging demand trends, station utilization behavior, and temporal patterns to generate adaptive pricing strategies for sustainable EV infrastructure management.

GitHub Repository:
https://github.com/satyam9140/EV_DYNAMIC_TARIFF_OPTIMIZATION.git

Introduction

The rapid growth of Electric Vehicles (EVs) has significantly increased the demand for efficient and intelligent charging infrastructure. Traditional fixed pricing models fail to adapt to varying charging demand, leading to congestion, inefficient resource allocation, and revenue losses. This project proposes an Agentic AI-based dynamic tariff optimization framework capable of adjusting charging prices in real time according to network conditions, utilization rates, and demand patterns.

Objectives

The primary objectives of this project are:

• Analyze EV charging demand patterns over time.
• Detect overloaded charging stations and utilization bottlenecks.
• Design dynamic tariff strategies using intelligent optimization methods.
• Improve revenue generation while reducing congestion.
• Encourage off-peak charging behavior using adaptive pricing.
• Build visual analytics dashboards for decision support systems.

Methodology

The project workflow begins with collecting EV charging transaction data and preprocessing the dataset for analysis. Temporal features such as hour, weekday, lag demand, rolling demand averages, and occupancy trends are extracted. Machine learning-driven decision logic is then used to identify congestion levels and dynamically modify pricing strategies.

Dynamic pricing decisions are influenced by:
• Demand volume
• Charging duration
• Station occupancy
• Queue length proxy
• Utilization rate
• Peak hour detection
• Historical charging patterns

System Performance

The proposed framework achieved strong operational improvements in EV network management. Key performance indicators obtained from the analysis include:

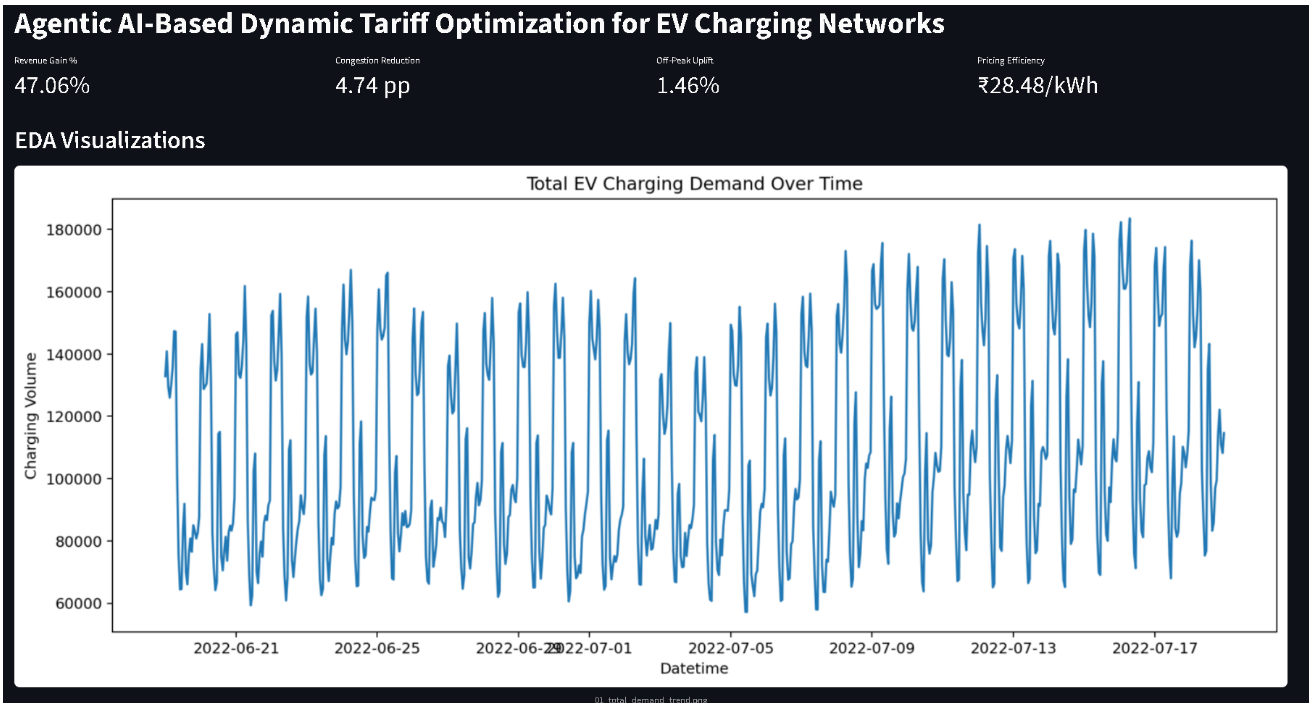
• Revenue Gain: 47.06%
• Congestion Reduction: 4.74 percentage points
• Off-Peak Uplift: 1.46%
• Pricing Efficiency: ■28.48/kWh

These results demonstrate that intelligent tariff adaptation can significantly improve charging infrastructure efficiency while balancing user demand.

Data Visualization and Analysis

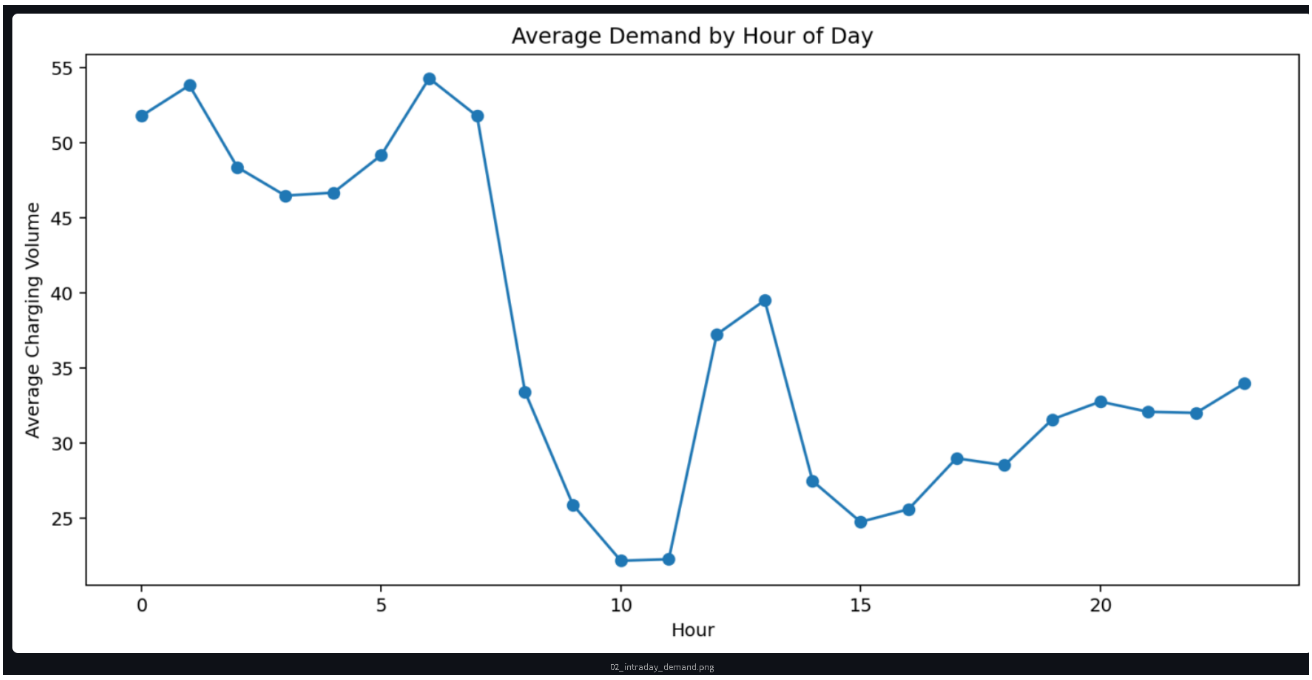
Several analytical visualizations were generated to understand charging behavior, temporal demand variation, utilization trends, and overloaded station analysis. These plots help identify critical operational insights and support intelligent pricing optimization decisions.

Total EV Charging Demand Over Time



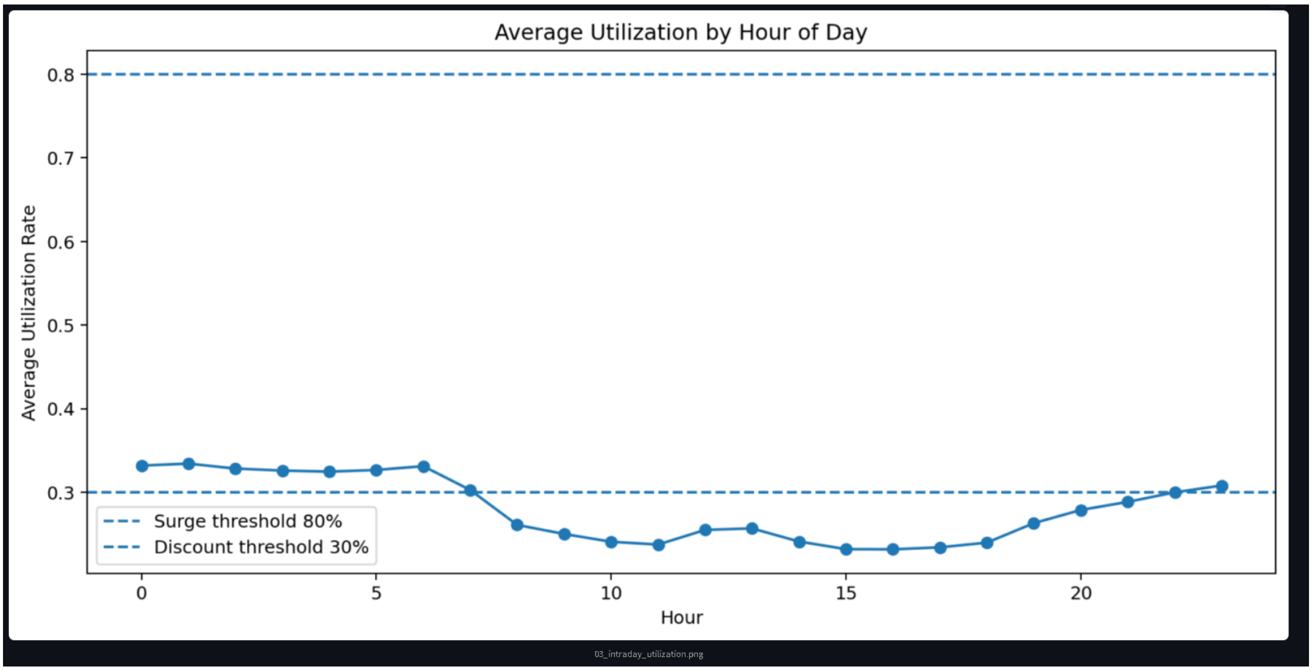
This visualization represents the total EV charging demand variation over time. Strong cyclical demand fluctuations can be observed, indicating peak usage periods and high charging intensity during specific intervals. Such trends are essential for implementing adaptive pricing models.

Average Demand by Hour of Day



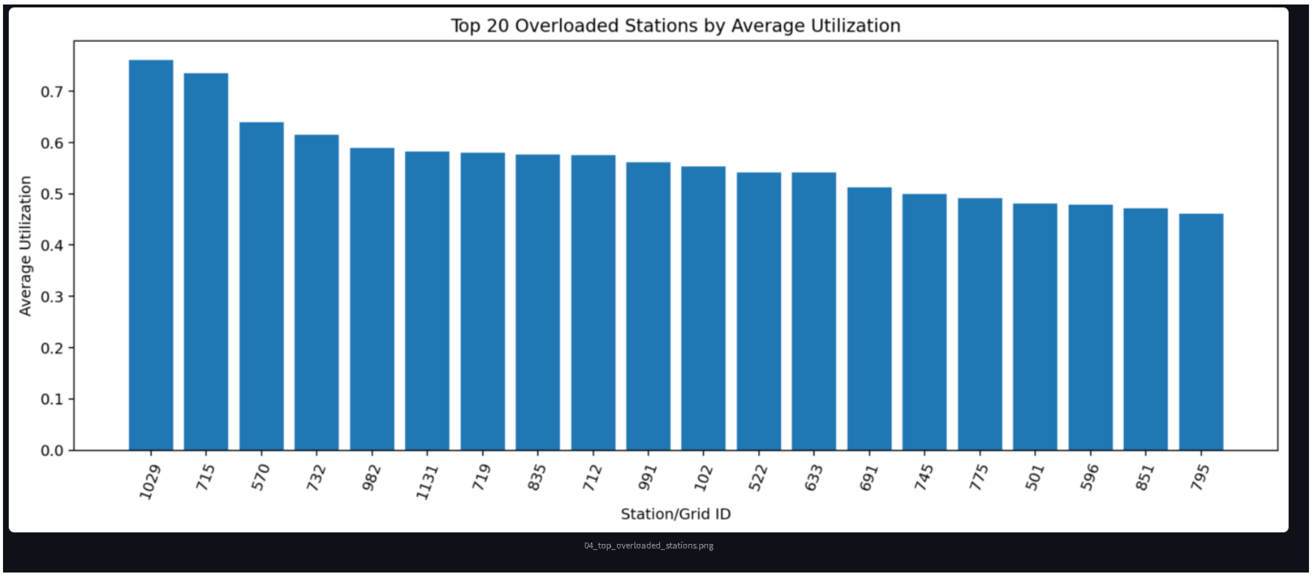
This graph shows the average charging demand by hour of the day. Early morning and evening periods exhibit relatively higher charging demand, while midday demand remains comparatively lower. This information helps optimize tariff scheduling.

Average Utilization by Hour of Day



The utilization analysis demonstrates average station occupancy trends throughout the day. The plotted thresholds indicate operational limits for surge pricing and discount pricing mechanisms. Utilization-based dynamic pricing helps reduce station overload.

Top Overloaded Stations by Average Utilization



This bar chart identifies the most overloaded charging stations within the network. Stations with high utilization rates require optimized load balancing, infrastructure expansion, or surge pricing to control congestion.

Dynamic Tariff Decisions Table

Dynamic Tariff Decisions

	station_id	hour_datetime	demand_volume	avg_occupancy	avg_duration	avg_raw_price	utilization_rate	queue_length_proxy	capacity	hour	dayofweek	is_weekend	is_peak_hour	CBD	dynamic_pricing	lag_1h_demand	lag_24h_demand	rolling_3h_demand
0	1000	2022-07-13 05:00:00	315	68	5.25	0.9049	0.3523	0	193	5	2	0	0	0	0	314.51	342.51	321.3149
1	1000	2022-07-13 06:00:00	308.5717	66.1667	5.1155	0.9049	0.3428	0	193	6	2	0	0	0	0	315	332.9122	317.8149
2	1000	2022-07-13 07:00:00	269.9356	57.5	4.3993	0.9049	0.2979	0	193	7	2	0	0	0	0	308.5717	284.376	312.6939
3	1000	2022-07-13 08:00:00	211.8239	46.1667	3.4533	0.9049	0.2392	0	193	8	2	0	1	0	0	269.9356	206.8519	297.8357
4	1000	2022-07-13 09:00:00	170.2983	38.0833	2.7611	0.9049	0.1973	0	193	9	2	0	1	0	0	211.8239	137.3847	263.4437
5	1000	2022-07-13 10:00:00	137.3594	32.25	2.2798	0.9049	0.1671	0	193	10	2	0	1	0	0	170.2983	106.3767	217.3526
6	1000	2022-07-13 11:00:00	137.4061	32	2.2608	0.9049	0.1658	0	193	11	2	0	0	0	0	137.3594	119.3801	173.1606
7	1000	2022-07-13 12:00:00	152.1236	35.4167	2.5294	0.9049	0.1835	0	193	12	2	0	0	0	0	137.4061	142.0708	148.3546
8	1000	2022-07-13 13:00:00	150.2997	35.5	2.5393	0.9049	0.1839	0	193	13	2	0	0	0	0	152.1236	155.3728	142.2964
9	1000	2022-07-13 14:00:00	147.246	44.1667	2.486	0.9049	0.2288	0	193	14	2	0	0	0	0	150.2997	134.4544	146.6098

The dynamic tariff decision table presents the engineered dataset used by the AI-based pricing engine. Features such as occupancy, utilization rate, rolling demand, lag demand, and charging duration collectively influence pricing decisions.

Performance Metrics Summary

Metric	Value
Revenue Gain	47.06%
Congestion Reduction	4.74 pp

Off-Peak Uplift	1.46%
Pricing Efficiency	■28.48/kWh

Conclusion

This project successfully demonstrates the application of Agentic AI techniques for dynamic tariff optimization in EV charging networks. By analyzing charging demand, utilization patterns, and congestion trends, the proposed system dynamically adapts pricing strategies to improve operational efficiency and revenue generation.

The framework supports sustainable EV infrastructure management by balancing charging demand across peak and off-peak hours. Future improvements may include reinforcement learning-based tariff optimization, real-time IoT integration, predictive congestion forecasting, and smart grid interaction for enhanced scalability.

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